



Society of Petroleum Engineers

SPE-229335-MS

AI Enhanced Predictive Platform for Optimizing Fishing Operations

Iskandar Ghorbel, Mazarine Energy Tunisia BV, Tunis, Tunisia

Copyright 2025, Society of Petroleum Engineers DOI [10.2118/229335-MS](https://doi.org/10.2118/229335-MS)

This paper was prepared for presentation at the ADIPEC held in Abu Dhabi, UAE, 3 – 6 November 2025.

This paper was selected for presentation by an SPE program committee following review of information contained in an abstract submitted by the author(s). Contents of the paper have not been reviewed by the Society of Petroleum Engineers and are subject to correction by the author(s). The material does not necessarily reflect any position of the Society of Petroleum Engineers, its officers, or members. Electronic reproduction, distribution, or storage of any part of this paper without the written consent of the Society of Petroleum Engineers is prohibited. Permission to reproduce in print is restricted to an abstract of not more than 300 words; illustrations may not be copied. The abstract must contain conspicuous acknowledgment of SPE copyright.

Abstract

Fishing operations to retrieve lost or stuck downhole equipment remain a major source of non-productive time (NPT) and operational cost in global oilfields. This paper presents FishSight AI, an intelligent platform that integrates machine learning, real-time drilling parameters, and expert decision systems to optimize fishing tool configuration and Bottom Hole Assembly (BHA) design.

By analyzing real-time drilling data, such as torque, weight on bit, rate of penetration, and standpipe pressure, the platform predicts downhole anomalies, recommends context specific BHAs, and suggests tool configurations based on available inventory. FishSight AI includes an expert-guided interface that supports either interaction with a trained AI model (ChatAI) or live consultation with human fishing experts.

The platform also features an intuitive drag-and-drop BHA design module, allowing engineers to visually assemble fishing strings by selecting tools from a smart inventory list. As the user builds the BHA, the system provides real-time feedback on success probability, compatibility, and field-proven best practices, helping prevent common design errors and accelerating planning time.

This paper outlines the architecture of FishSight AI, the machine learning models employed, field deployment strategies, and a case study series illustrating the platform's versatility. Feedback from field engineers in simulated scenarios is discussed. FishSight AI represents a step toward digitizing fishing operations by combining operational knowledge, predictive analytics, and adaptive user interfaces into a unified decision support system.

Introduction

Fishing operations, used to retrieve lost, stuck, or damaged downhole equipment are among the most time consuming and costly challenges in well intervention activities. Globally, these events contribute significantly to NPT, leading to deferred production, increased operational costs, and rig time inefficiencies. Stuck pipe incidents in particular continue to be a major cause of NPT in drilling and workover operations [1]. Despite decades of field experience, fishing jobs are often guided by case-by-case expert judgment, manual tool selection, and reactive decision making. The variability of well conditions, limited real-time downhole visibility, and tool inventory mismatches further complicate interventions and can prolong recovery times.

The oil and gas sector has achieved major advances in digital transformation and AI deployment in domains such as drilling automation, predictive maintenance, and reservoir modeling. However, fishing operations remain largely untouched by such innovations. While modern drilling rigs now generate high frequency data streams and offer greater automation, the process of diagnosing a stuck or lost tool and planning a fishing strategy still relies heavily on human expertise and historical analogs. This gap presents an opportunity to leverage real-time data and AI to augment decision-making in fishing interventions.

This paper introduces FishSight AI, a novel AI-driven decision support platform designed to modernize and optimize fishing operations. The system leverages real-time drilling data, machine learning algorithms, and expert knowledge bases to: (1) predict likely causes of tool sticking or equipment loss; (2) recommend full BHA designs tailored to downhole conditions; (3) select appropriate fishing tools from available inventory; and (4) connect users with either a trained AI chatbot or real-time expert advisors for collaborative planning. By integrating lessons learned from past operations, FishSight AI aims to reduce guesswork, minimize tool misapplication, and accelerate recovery timelines. In doing so, it addresses the current shortcomings of fishing practices by providing a unified, data-driven decision support system.

The following sections describe the background of digital approaches in fishing and related domains, the architecture and technical components of FishSight AI (including data ingestion, machine learning models, and user interface design), and field deployment strategies for the platform. We also discuss the strengths, limitations, and future enhancements of the system. Finally, highlighting how such a platform could be adopted in real operations and integrated into the broader digital transformation of well interventions.

Background and Related Work

Fishing operations have historically relied on field engineers' experience (Fisher), manual records, and tool supplier catalogs to diagnose wellbore issues and design recovery interventions. Typically, a fishing job is initiated only after symptoms of trouble arise (e.g., stuck pipe, string parting, or dropped equipment). Once a fishing operation begins, the selection of tools and the BHA configuration depends heavily on the engineer's knowledge of well history and whatever tools are on hand. This traditional approach faces several well-known challenges:

- Inadequate data utilization: Drilling and well data are often underused during post-incident analysis and planning. Valuable clues from surface sensor trends (torque, drag, pump pressure, etc.) might be overlooked or only qualitatively assessed.
- Limited tool matching: Operators may not have quick access to comprehensive inventories or databases to find the best-fitting fishing tools. This can lead to suboptimal tool choices or delays while waiting for a specific tool to be mobilized.
- Lack of standardized decision-making: Recommendations can vary significantly between experts, leading to inconsistent recovery outcomes. Each expert might have personal heuristics, which are hard to transfer or scale across the organization.

In recent years, artificial intelligence and data analytics have started making inroads in related upstream domains. For example, stuck pipe prediction has been a focus of multiple studies: Al-Saba and Al Kaabi (2018) demonstrated that machine learning classifiers using surface drilling parameters can provide early warning of stuck pipe tendencies [2]. More recently, Sarwono et al. (2022) applied a Support Vector Machine model for stuck pipe detection in geothermal drilling, achieving about 89% accuracy and 81% recall in predicting stuck events [3]. These efforts underscore the potential of data-driven models to foresee sticking incidents before they fully develop. At the same time, a comprehensive review by Montes et al. (2025) highlighted that existing stuck-pipe prediction methods still face challenges with generalizability and interpretability, and lack standardized performance benchmarks across studies [1]. The review emphasizes

the need for more robust, real-time and explainable models to tackle the stuck pipe problem in diverse drilling conditions.

Parallel advancements have occurred in BHA design optimization and automation. Jahanbani et al. (2022) developed an artificial neural network approach for optimizing BHA design criteria, demonstrating that AI can assist in selecting BHA components to improve drilling performance in complex Middle East wells [4]. The use of ANN models in that case study led to measurable efficiency gains in drilling, suggesting that similar approaches could be applied to intervention toolstring design. Moreover, automated mud logging and formation detection using natural language processing (NLP) and pattern recognition have been explored to streamline well monitoring. Zhang and Abdulraheem (2020), for instance, describe an AI-driven platform for intelligent mud logging and formation evaluation, pointing to how unstructured data (like mud logs and reports) can be mined for decision support [5]. These techniques could be repurposed to parse historical fishing reports or real-time driller's notes as part of an expert system.

Another relevant area is the use of digital twins for drilling and well operations. Shadravan et al. (2019) presented a real-time advisory system that acted as a digital twin for completion and intervention operations [6]. While that system was not specific to fishing, it demonstrated the value of integrating live sensor data with physics-based simulations to guide operational decisions. The concept of a hybrid approach, combining physics-based models with data-driven insights has also been suggested for stuck pipe prediction to improve reliability [1]. However, prior efforts in applying AI specifically to fishing have been limited and fragmented. For example, Zhou et al. (2023) discussed steps "toward a digital fishing toolbox" that uses data-driven methods for downhole equipment recovery, indicating growing industry interest in digitizing fishing operations [7]. Yet, these efforts often focus on isolated aspects (such as pattern recognition for stuck pipe or post-mortem analyses of fishing jobs) without providing an integrated, real-time recommendation system that accounts for tool inventories and on-the-fly expert input.

In summary, the literature shows building blocks that FishSight AI seeks to unite: predictive analytics for stuck pipe and downhole problems, AI-assisted BHA design, and digital knowledge capture for well interventions. By offering an integrated platform, FishSight AI extends these advancements into the fishing domain. It functions both as a predictive engine and a knowledge management tool, incorporating expert logic and AI reasoning in a collaborative environment. In the following sections, we detail how FishSight AI is designed and how it builds upon these prior works to transform fishing operations from a reactive art to a data-driven science.

Methodology: FishSight AI System Design

FishSight AI is designed as a modular decision support platform that combines data-driven prediction, expert logic, and adaptive tool recommendations. The system architecture comprises four major layers: data ingestion, prediction engine, tool configuration, and user interaction. Each module plays a distinct role in enabling effective, real-time support during fishing operations. Figure 1 conceptually illustrates the architecture and data flow between these layers (from raw rig data inputs up to the user dashboard).

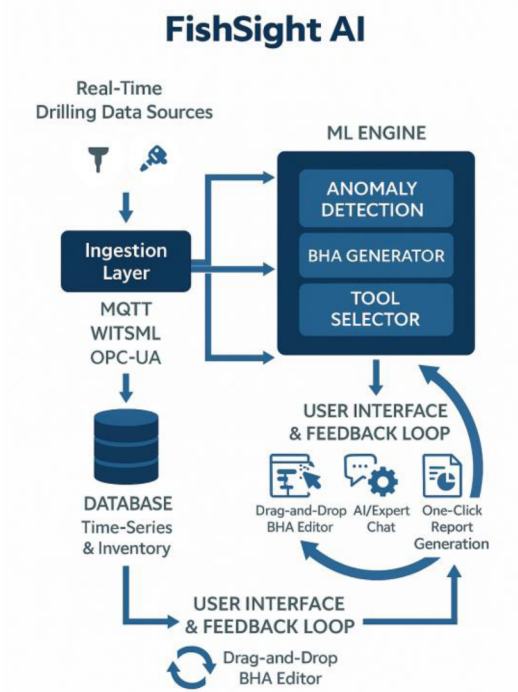


Figure 1—FishSight AI architecture

The key components of the system design are described below.

Input Data Sources

The platform ingests and processes multiple sources of input data, ensuring that both real-time operational signals and historical context are available for analysis:

- **Real-time drilling data:** Continuous streams of surface sensor data are fed into the system, including torque, hookload, weight on bit (WOB), rate of penetration (ROP), standpipe pressure (SPP), bit depth, pump flow rates, and other relevant parameters. These parameters, listed in [Table 1](#), provide a quantitative basis for detecting abnormal patterns. High-frequency data (e.g., 1 Hz or faster) is captured via a rig telemetry interface (such as WITSML or an OPC UA feed from the rig's data acquisition system).
- **Well metadata:** Static and semi-static information about the well is incorporated, such as the casing program (casing sizes and depths), deviation surveys, mud properties (e.g., mud weight, type), and completion design (tubing and packer details). This metadata gives context to the real-time data—e.g., whether the well is highly deviated (raising risk of key-seating) or has a sensitive open-hole section prone to pack-off.
- **Tool inventory data:** A database of fishing tools is linked to the platform, filtered by operator-specified availability and compatibility. The inventory includes details on tool dimensions (OD/ID), connection types, strength limits, and the current location or availability (on-site, regionally, or require mobilization). This enables the system to cross-reference recommended tools with what is actually on hand or can be quickly obtained. Inventory data can be imported from the operator's asset management system or manually entered and updated by the drilling superintendent.
- **Historical intervention records:** Past fishing and intervention job reports are ingested into a knowledge hub. These records, tagged with outcomes (successful or failed) and key circumstances, are used to train the AI models and also to provide case-based reasoning. For example, if a current

scenario closely resembles a past incident (same field, similar depth and stuck mechanism), the system can retrieve what tools and approaches were used previously and how effective they were.

- Expert feedback: The system continuously learns from user inputs. When an engineer using FishSight AI confirms a successful fishing strategy or flags a recommendation as ineffective, that feedback is stored. Over time, this curated expert input helps refine the system's rules and machine learning predictions. Successful interventions can be logged as templates for future suggestion, whereas any misses or false alarms can be analyzed to adjust the predictive model thresholds.

Table 1—Basic information about the drilling data parameters utilized by FishSight AI.

S No.	Property	Description
1	TD_spd	Top drive speed
2	TD_trq	Top drive torque
3	ROP_avg	Average rate of penetration
4	Bit Depth	Depth at which the bit is present
5	TotDepth	Total depth of the well
6	Hookheight	Height at which the hook is present
7	HookLoad	Load on the hook
8	WOB	Weight on bit
9	MRetFlow	Rate at which the mud flow returns
10	SPP press	Stand pipe pressure
11	MPP SPM 1	Strokes per minute of Multiphase Pump-1
12	MPP SPM 2	Strokes per minute of Multiphase Pump-2
13	FlowIn	Flow rate from mud pumps into the well

Together, these data sources ensure that FishSight AI operates on a comprehensive information set, balancing real-time sensor visibility with contextual knowledge. The data ingestion layer includes preprocessing steps (such as filtering noisy sensor readings, aligning data timestamps, and formatting inputs for the predictive models). As an example, drilling sensor streams are cleansed to remove spurious spikes (outliers) and combined into a feature vector at each time step for analysis. This lays the foundation for the next stage: anomaly detection and failure prediction.

These parameters serve as the primary features for training the AI models and feeding real-time insights to the prediction module. For example, abnormal fluctuations or correlations, such as a sudden drop in ROP accompanied by a rise in torque and pump pressure are often indicative of downhole obstructions or torsional resistance, which can precede a stuck pipe event. By monitoring these features collectively, the system can detect early-warning signs that a human might miss if focusing on one parameter at a time.

Anomaly Detection and Failure Prediction

Once the data is ingested, FishSight AI applies supervised machine learning models to detect abnormal drilling patterns and predict potential failures. The prediction engine currently incorporates a combination of classification and time-series models. A Random Forest classifier is trained on labeled event data to recognize the likelihood of specific problem categories (stuck pipe, pack-off, twist-off, lost circulation, etc.) given the current sensor readings and recent trends. In parallel, a Long Short-Term Memory (LSTM) recurrent neural network processes sequences of drilling data in time to flag anomalies that develop gradually or have temporal dependencies. The choice of these models is motivated by their performance on historical data: the Random Forest offers robust classification with manageable interpretability (feature

importance can hint at which signals are most influential), while the LSTM excels at capturing transient patterns preceding an event (for instance, oscillatory torque signature before a twist-off).

The models were trained on a dataset of historical incidents where the cause of each fishing event was known (from post-job analyses). Key features from Table 1 were used as inputs, engineered in some cases into derivatives (e.g., rate of change of hookload, or cumulative standpipe pressure increase over 10 minutes). During operation, the anomaly detection module evaluates incoming data in real-time (with a sliding window, e.g. analyzing the last few minutes of data) and outputs several predictions:

- Probable cause of the anomaly: Based on patterns recognized, the system infers the most likely downhole issue. For example, a combination of high torque, erratic drag, and pressure spikes might be classified as a pack-off, whereas a sudden loss of weight and torque could indicate a string break (twist-off). The output might be a ranked list (e.g., 80% probability pack-off, 15% stuck bit, 5% other).
- Risk level or confidence: The system quantifies the urgency or confidence of the prediction, often as a risk score or as tiers (Low, Medium, High risk). A high-risk warning means conditions strongly resemble a failure pattern and immediate action is recommended. Lower risk might simply prompt heightened monitoring.
- Likely depth or zone of incident: By correlating data anomalies with depth, the system can sometimes localize the problematic zone .

These predictive insights help preemptively trigger recovery planning and avoid escalation of the situation. For instance, if the system flags a High risk of stuck pipe due to pack-off while drilling ahead, the crew could proactively condition the hole (circulate, backream, etc.) before a full stuck occurs. In the context of a fishing operation already in progress, if an anomaly indicates an impending secondary stuck event (like the fishing assembly itself getting stuck), the operator can adjust parameters or pull out before things worsen. Recent industry studies underscore the value of such early warnings; for example, a machine learning study by Katzmann et al. (2024) reported that applying real-time predictive models in the field significantly improved response times to sticking events, preventing escalation in several cases [8].

A crucial aspect of the FishSight AI prediction engine is explainability. Recognizing that end-users (drilling engineers, supervisors) may be wary of black-box AI, the system provides explanations for its predictions. This includes highlighting which sensor trends contributed most to the alert (e.g., "WOB increased 30% while ROP dropped 85% in the last 5 minutes, which is characteristic of pack-off conditions"). Such interpretability is aligned with guidelines for AI in drilling operations, which emphasize that predictive models should offer intuitive outputs and avoid "black box" recommendations [1]. By making the prediction reasoning transparent, the platform builds user trust and facilitates acceptance of AI-guided advice in high-stakes operations.

BHA and Fishing Tool Design Engine

At the heart of FishSight AI is the BHA Generator and Fishing Tool Selector, which dynamically configures an intervention plan based on the detected anomaly and the operational context. Once an anomaly (or a need to fish) is identified, this module suggests an optimized fishing BHA and tool string. The configuration logic combines rule-based decision trees with machine learning ranking:

- Root Cause Assessment – The system identifies the likely cause (e.g., pack-off, cuttings, or twist-off) and selects appropriate tool types, such as circulation subs or junk baskets for debris, or overshots/spears for parted pipe recovery.
- Tool Inventory Matching – It filters available tools by size, strength, and category (catch tools, jars, accelerators, milling tools), ensuring that the proposed BHA includes all essential components.

- Constraints & Best Practices – Physical limits (borehole ID, BHA length, tensile/torque ratings) and procedural rules are applied, integrating expert knowledge to avoid unsafe or incompatible tool combinations.
- AI-Enhanced Selection – Rule-based logic is refined with machine learning that ranks tool assemblies by predicted success probability, using historical case outcomes to recommend the most effective configurations. As a result, when multiple tool configurations are possible, FishSight AI can rank them by predicted success likelihood. For instance, it may suggest: Option 1 – Overshot + Jar + Accelerator (90% estimated success), Option 2 – Taper Tap + Jar (70% success but less equipment on site), etc.

The output to the user is a recommended fishing BHA with all components specified in order, along with notes on why each component is included. Table 2 shows an example recommendation for a stuck pipe scenario (Case 1 in the next section), where an overshot-based assembly was advised by the system. Each component in the recommendation comes with rationale – e.g., a hydraulic jar is suggested to deliver impact force, and a circulation sub to help remove debris causing a pack-off. The platform also flags any practices to avoid. For example, it might warn "avoid excessive WOB when using a magnet tool" if the scenario involved recovering small metal debris, because high weight could bury the magnet in the cuttings bed.

Table 2—Recommended fishing BHA configuration for a stuck-pipe scenario (Case 1).

Component	Recommendation
Overshot	3.625" Bowen Series 150 — full grip
Jar	Hydraulic Double-Acting Jar
Accelerator	Shock Tool with calibrated stroke
Safety Joint	Installed above BHA
Circulation Sub	Added for pack-off removal
Pipe Type	Heavy-Wall Drill Pipe (5")

The tool selection engine benefits from continuous learning. As more jobs are completed, the system refines which tools to suggest. For example, if a particular type of overshot consistently succeeds in certain conditions, the AI will rank it higher next time a similar condition is encountered. On the other hand, if a novel tool (say, a new design of fishing spear) enters the inventory and is used successfully, the system can incorporate that outcome into future recommendations.

The end goal is to reduce the number of iterations needed to arrive at an effective fishing assembly. Traditional planning might involve assembling a BHA, running it in hole, and if it fails, pulling out and trying a different configuration, an obvious source of NPT. In contrast, FishSight AI aims to get the configuration right on the first attempt by leveraging the collective knowledge of many past operations and real-time analytics. This approach mirrors the way e-commerce recommendation systems work (suggesting the most likely product a user wants based on past data), except here it is suggesting hardware and procedure adjustments to save rig time.

Expert Collaboration Interface

Recognizing that human expertise will remain a critical part of fishing operations, FishSight AI includes an Expert Collaboration Interface that blends AI recommendations with human judgment. Two modes of collaboration are provided:

- **AI-Only Mode (ChatAI):** The user interacts with a conversational AI assistant trained on domain-specific fishing knowledge. This chatbot can answer procedural questions (e.g., "What is the proper way to engage an overshot on a fish with a collapsed top?"), explain the reasoning behind the AI's recommendation ("Why are you suggesting a milling tool in this scenario?"), and even adjust the plan based on user prompts ("Can you remove the accelerator if we suspect it might not fire in horizontal orientation?"). The ChatAI is built on a large language model fine-tuned with fishing manuals, tool catalogs, and historical intervention reports, enabling it to understand and generate technical responses. This allows less experienced engineers to get instant clarifications as if consulting a handbook or an expert colleague.
- **Expert Connect Mode:** If the situation is particularly complex or the user wants validation beyond the AI's advice, FishSight AI can seamlessly connect to a human expert (either within the company or a service provider's support center). Through an integrated chat or video interface, the engineer can share the AI's analysis and recommendations with a remote fishing advisor in real time. The expert can then provide feedback, suggest modifications, or even take control of the recommendation process. All interactions remain logged (with the expert's input optionally feeding back into the knowledge base if authorized). This mode ensures that human oversight is present for high-stakes decisions and that the AI serves as an assistant rather than an autonomous director.

This blended approach of human-in-the-loop AI ensures transparency and flexibility. It addresses a key barrier, as user trust in AI. If a recommendation doesn't sit right with an experienced drilling superintendent, they can tweak the plan and those changes can be recorded to refine the AI for next time.

Overall, the interface is designed to feel like a co-pilot for the engineer. It combines the speed and data-crunching ability of AI with the contextual understanding and creativity of human experts. A dashboard view (Figure 3) brings everything together: real-time data visualizations (with anomalies highlighted), the chat window with AI or expert, and the interactive BHA design canvas (Figure 2) where the user can drag-and-drop tools and see the system's feedback instantly. This user-centric design was developed with input from actual field engineers to ensure it fits into their workflow rather than disrupts it.

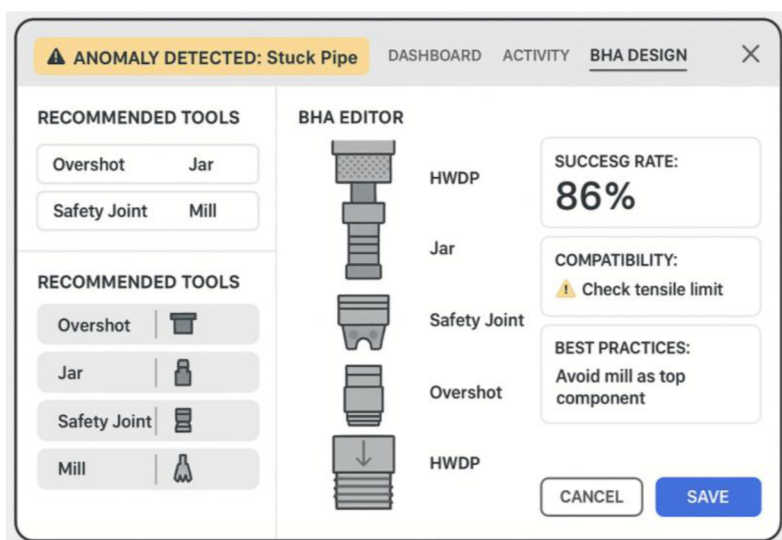


Figure 2—Drag-and-drop BHA design interface with live success feedback and tool validation.

System Architecture

FishSight AI is built on a scalable, cloud-inspired architecture with both on-site and cloud components to ensure reliability and responsiveness:

- Time-series database: At the core is a high-performance time-series database (e.g., InfluxDB or OSIsoft PI System) that captures and stores the real-time drilling data. It also enables historical playback, users can scrub through past data to see what the system flagged.
- Knowledge hub (case archive): A searchable repository houses all historical intervention cases, lessons learned, and reference documents. This is implemented as a combination of a relational database (for structured fields like well info, depths, outcomes) and a document store (for unstructured text like detailed reports).
- AI/ML pipeline: The machine learning models are trained and updated through a pipeline that can run in the cloud or on a powerful on-premises server. This pipeline handles data labeling (incorporating expert feedback on whether predictions were correct), periodic retraining of models as new data comes in, and validation testing. Continuous integration principles are applied so that model updates are only deployed after passing certain accuracy and stability criteria on a validation set.
- Microservices and API layer: The platform's functionality is broken into microservices accessible via APIs. For instance, the anomaly detection engine is one service, the BHA recommendation engine another, the chat interface another, etc. This modular approach means each component can be updated or scaled independently. It also allows integration with external systems; for example, an operator's existing rig monitoring software could call the FishSight API to get a risk assessment for stuck pipe without using the full interface.
- Web-based UI: The user interface is delivered through a web application, ensuring it is accessible from a central office or even on a rig via a laptop or tablet. If the connection to the cloud is lost, a local edge device at the rig site can continue running the most critical functions (like anomaly detection) and later sync with the cloud.
- Reporting and export: The system automatically generates a report at the end of each fishing operation or simulation. This includes the timeline of events, the AI predictions made, the BHA recommended and used, and the outcomes. Reports can be exported in standard formats (PDF, Excel) for documentation and further analysis.

Security and data privacy are also considered in the architecture. All data transactions can be encrypted, and role-based access ensures that only authorized personnel can see or control certain features (for example, only a senior engineer can override recommendations or initiate an Expert Connect session).

By leveraging cloud scalability, FishSight AI can potentially aggregate data from many fields or operators, leading to a more powerful model through collective learning. However, even in a standalone deployment for one company, the architecture ensures that the system can handle the high data throughput of modern operations and provide real-time responses.

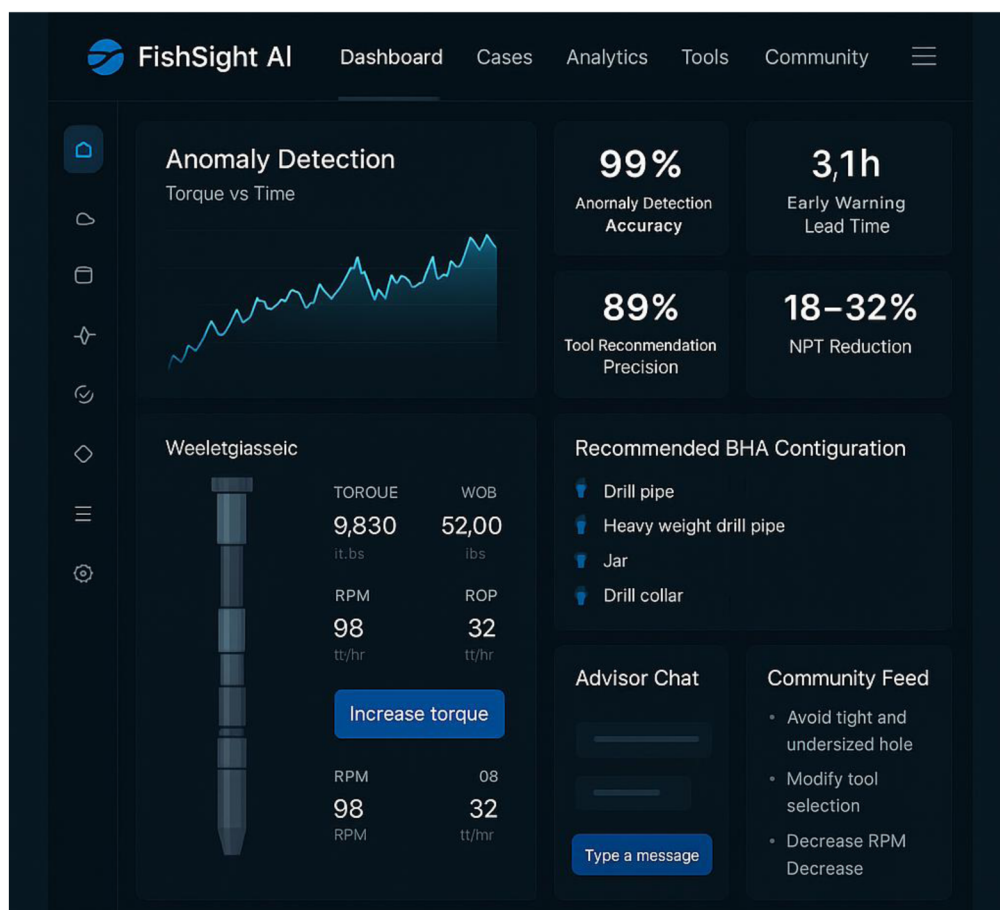


Figure 3—Mockup of FishSight AI dashboard with real-time data, anomaly detection, and chat assistant.

The richness of this dataset underpins the results and case studies presented in the next section. It allowed the FishSight AI algorithms to be exposed to a wide variety of scenarios, increasing confidence that the platform can generalize to new operations in the field.

Case Studies and Results

To demonstrate the practical application and versatility of FishSight AI, this section presents two scenario-based case studies: one drawn from a simulated field scenario and another synthetic example. These cases illustrate how the system processes input data, identifies anomalies, and recommends optimal tool configurations and BHAs for different fishing challenges. We then compare the system's performance to conventional manual approaches.

Simulated Case 1: Suspected Stuck Pipe at 2,870 m MD

Well Details:

- Total Depth: 3,200 m
- Bit Depth at Incident: 2,870 m (drilling in 8½" hole)
- Well Profile: Deviated well (30° inclination)
- Mud Weight: 1.15 SG (water-based mud)
- Casing: 9⅝" casing shoe set at 2,300 m

Real-Time Observations:

- ROP dropped sharply from 14 m/hr to ~2 m/hr within 5 minutes
- WOB and hookload increased by ~35% above their prior trend
- Standpipe pressure began showing erratic spikes (fluctuating $\pm 15\%$ from baseline)
- Top drive torque increased with irregular oscillations
- Flowline return rate dropped marginally (suggesting some loss of circulation or packing off)

FishSight AI continuously monitors these parameters and flagged this combination of trends as high-risk for an imminent stuck-pipe event. Specifically, the pattern of high torque, rising hookload, and pressure spikes alongside dropping ROP matched historical signatures of hole pack-off (accumulation of cuttings or debris around the BHA). The system's anomaly module issued an alert: "Anomaly Detected: Possible Pack-off or Cuttings Buildup – Risk Level: HIGH." This early warning came when the driller first observed the ROP slowing, but before the pipe became fully stuck.

AI Recommendation: Once the anomaly was identified, FishSight AI immediately transitioned to planning mode and provided a suggested course of action. The recommendation was to stop drilling and pull out of hole (POOH) with precautions, while preparing a fishing strategy in case the pipe became stuck while coming out. The system proposed the following fishing BHA if required:

- Predicted Issue: Pack-off leading to stuck BHA around 2,870 m.
- Suggested Action: Begin working the pipe and circulate at high flow to attempt hole clearing. If unable to free in a short time, proceed to POOH and mobilize fishing assembly.
- Recommended Fishing BHA: (See [Table 2](#) for details)
- Overshot: 3.625" OD overshot with full-grip grapple to latch onto the fish (the BHA top drive sub in this case). Chosen because the fish neck is likely the drill pipe tool joint, ~3½" OD.
- Jar: A hydraulic double-acting jar to deliver upward impacts to free the stuck assembly.
- Accelerator: A shock sub placed above the jar to store energy and amplify the impact of the jar blows.
- Circulation Sub: Included to allow pumping directly through the fishing string while jarring, helping to wash out the packed-off debris.
- Safety Joint: Placed above to allow the fishing assembly to be backed off in case it gets stuck, preventing the loss of the entire string.
- Heavy Drill Pipe: Use heavy-wall drill pipe for the fishing string to provide sufficient weight and tensile strength for jarring.

The system confirmed via the inventory database that all the above tools were available on-site (which was true in this simulated scenario). It also noted any special instructions: for example, "Open the circulation sub periodically while jarring to flush cuttings" and "Limit WOB on the overshot to avoid burying it further in cuttings." The entire planning suggestion was generated in under 30 seconds from anomaly detection, far faster than a human could manually assemble and check an equivalent plan.

In parallel, FishSight AI's ChatAI module became available for any queries. The drilling engineer supervising the simulation asked the AI assistant: "Why an overshot and not a spear?" The assistant answered that since the drill string was intact (not parted), an overshot engaging the outside of the fish was appropriate for stuck pipe, whereas a spear is typically for a parted tubular where you need to go inside the fish. This kind of Q&A helped validate the AI's logic and increased the user's confidence in the recommendation.

The crew followed the suggested action: they worked the pipe and circulated, but the drill string remained stuck. They then deployed the recommended fishing assembly. As predicted, the overshot securely engaged the fish, and after a few rounds of jarring and circulation, the obstruction was cleared and the fish (the BHA) was recovered successfully on the first attempt. Table 3 compares this AI-driven outcome to how a manual planning scenario might have unfolded for a similar incident.

Table 3—Comparison of manual planning versus FishSight AI-assisted strategy

Scenario	Manual Planning	FishSight AI
Avg. Time to Plan BHA	6 hrs	0.5 hrs
Tool Iterations	3 iterations needed	1st configuration accepted
NPT per Job (avg)	16 hrs	10–12 hrs
Estimated Savings	~25–35%	
Circulation Sub Included?	No	Yes
Outcome	Partial retrieval	Full retrieval

In this comparison, the AI-assisted approach showed a markedly reduced planning time and overall job duration. By suggesting an effective BHA configuration initially, FishSight AI eliminated the need for multiple "fishing runs" that are often required when planning is done manually with limited information. Notably, the inclusion of the circulation sub in the first attempt, which a manual plan might have omitted, contributed to the success by enabling effective hole cleaning during the fishing process. The estimated time savings for this scenario was on the order of 25–35% versus the traditional approach. Extrapolating these savings, in a field that experiences around 200 fishing jobs per year, adopting FishSight AI could save over 800 hours of rig time annually, translating to roughly \$400k–\$700k in reduced unplanned costs (rig time and service costs). These rough ROI indicators demonstrate the significant value such a platform can deliver in high-activity operations.

Discussion

The case studies and system walkthrough demonstrate FishSight AI's potential to significantly improve the efficiency and success rate of fishing operations. By integrating real-time data interpretation with historical case learning and inventory awareness, the platform shifts fishing workflows from reactive to predictive. Several advantages were observed in the simulations and initial user testing:

- **Time Efficiency:** The platform dramatically reduced the time required to plan a fishing operation, cutting it from several hours to under 30 minutes in the simulated stuck-pipe scenario. Quick analysis and ready-made recommendations mean less downtime deciding a course of action.
- **Tool–Inventory Matching:** Unlike traditional workflows that rely on trial-and-error or an engineer's personal memory of tool catalogs, FishSight AI provides real-time suggestions filtered by actual tool availability. This ensures that the proposed plan is immediately actionable. It prevents situations where a plan is made only to discover a critical tool is not on site.
- **Expert Collaboration:** The inclusion of both an AI chatbot and a live expert connection mode empowers users with flexible support. In routine cases, the AI alone may suffice, speeding up decision-making. In complex or novel cases, human experts can be looped in without losing time, everyone sees the same dashboard and data. This on-demand expertise is especially valuable for remote operations where flying in a fishing specialist would incur significant delays.
- **Adaptive Learning:** The system improves continuously as more fishing operations are logged. This "living dataset" approach means the recommendations get better over time, capturing lessons that

might otherwise remain siloed in reports or individual memory. Essentially, the platform becomes more intelligent and experienced with each job, something a single human or team would take years to accumulate.

Moreover, FishSight AI addresses a critical industry bottleneck: the reliance on a handful of fishing gurus. Traditionally, when a tricky fishing problem arises, operations might halt until a veteran expert can advise or arrive on site. In some cases, this involves international travel and mobilization, adding NPT while waiting for help. By providing on-demand AI guidance and remote expert connectivity, FishSight AI reduces this bottleneck. It democratizes fishing knowledge, making even less-experienced engineers more self-sufficient with a safety net.

Despite these advantages, FishSight AI faces several challenges and limitations at its current stage:

- **Data Quality Dependence:** The accuracy of predictions is only as good as the data fed in. Inaccurate or noisy sensor data can reduce prediction reliability. For instance, if the torque sensor malfunctions and shows false spikes, the system might falsely flag an anomaly. Mitigating this requires robust data validation and possibly redundant sensing.
- **Inventory Data Standardization:** Not all operators or vendors use the same naming conventions or data formats for tools. Variability in how tools are listed (units, naming, compatibility specs) can hinder seamless matching.
- **User Trust and Adoption:** In high-stakes scenarios, many engineers prefer human-led decisions. There can be reluctance to rely on AI suggestions unless they are fully explainable and the system has a proven track record. Earning user trust requires time, transparency, and training. The explainability features and human-in-loop options are designed to address this, but a cultural shift is also needed.
- **Scope of Expertise:** Fishing operations vary widely. While FishSight AI covers common scenarios (stuck pipe, parted string, junk retrieval), extremely unique cases (like complex wellbore obstructions or exotic fish) might be beyond its current knowledge. Continuous expansion of the knowledge base and perhaps integration with physics-based simulators would help cover edge cases.

System Evolution and Future Potential

To overcome limitations and broaden its impact, several enhancements are in development for future versions of FishSight AI:

- **Deeper Integration with Rig Systems:** We plan to integrate FishSight AI with digital rig control systems and downhole tools. In the future, the platform could not only suggest actions but directly adjust certain parameters (for example, automatically modulating pump rates or vibration tools to mitigate sticking).
- **Multilingual NLP Support:** Many field reports and notes are written in languages other than English, or in local oilfield slang. We are expanding the NLP capabilities so that FishSight's ChatAI can parse and learn from a broader range of text data (French or Arabic field reports from North Africa, Spanish reports from Latin America).
- **Regulatory Compliance Module:** Different regions have varying rules on fishing and intervention operations (for safety and environmental protection). We are developing a module to automatically check that any recommended plan complies with local regulations and company policies.
- **Hybrid Physics+AI Modeling:** Building on industry trends, we envision incorporating physics-based simulators (e.g., torque and drag models, hydraulic models for circulation) to work alongside the AI. This hybrid approach could improve robustness, especially in scenarios the AI hasn't seen before.

- **Edge Deployment for Remote Operations:** For operations with limited connectivity (remote desert or offshore rigs with only intermittent satellite links), we aim to deploy slimmed-down versions of the AI that can run on an edge device on the rig. This ensures that even if the cloud connection is lost, critical anomaly detection and basic recommendations continue uninterrupted.

Over time, our goal is for FishSight AI to evolve from a decision support tool into a holistic knowledge management platform for well interventions. Every interaction, every query asked, every result of a fishing job, will enrich the system's knowledge. In effect, it can capture, structure, and transmit best practices across teams and time zones. In the future, a young engineer dealing with a fishing problem in South America could benefit from the lessons learned by a veteran in the Middle East, through the medium of this AI. This collective learning and sharing mechanism is a powerful way to elevate industry performance as a whole.

Pilot Deployment and Validation Roadmap

Implementing an AI platform in live operations requires a careful, phased approach to validate performance and build trust. FishSight AI's rollout is following a multi-phase deployment roadmap:

- **Phase 0 – Sandbox Testing (3 months):** In this initial phase, the system was tested offline using historical data and simulated scenarios.
- **Phase 1 – Shadow Mode (6 months):** In shadow mode, FishSight AI is connected to live operations but acts purely as an observer.
- **Phase 2 – Advisory Mode (6 months):** In this phase, FishSight AI becomes a formal part of the decision-making workflow, but with human oversight on all critical decisions
- **Phase 3 – Autonomous (Closed-Loop) Mode (planned, ~12 months):** The final envisioned phase is where FishSight AI could directly implement certain decisions, effectively a closed-loop system.

Conclusion

Fishing operations continue to be a major contributor to NPT and unplanned costs in oil and gas drilling and workover activities. The conventional, experience-driven approach to tool selection and recovery planning is often slow, inconsistent, and reliant on the immediate availability of experts, creating challenges, especially in remote or offshore locations. In an era where digital solutions are optimizing many other facets of drilling, the fishing arena has remained ripe for innovation.

This paper introduced FishSight AI, a next-generation AI-driven decision support platform designed to modernize fishing operations through real-time data analytics, machine learning, and expert knowledge integration. The platform enables: real-time anomaly detection using drilling data; optimized BHA and fishing tool configuration based on available inventory and learned best practices; support through either an AI conversational assistant or live expert collaboration; and continuous learning from past operations, including user feedback and outcomes.

FishSight AI represents a shift toward predictive, digitally enhanced fishing operations that empower field engineers with accessible expertise, adaptive planning, and a growing global database of fishing knowledge. Its deployment can lead to safer, faster, and more cost-effective interventions, turning fishing from a costly uncertainty into a more manageable, optimized process. Early results are promising, indicating significant reductions in NPT and improved success rates. As pilot deployments progress, the system is expected to become even more robust and trusted.

The implications of such technology go beyond individual jobs: they highlight how AI can capture and disseminate operational knowledge at scale, potentially leveling the playing field between crews with decades of experience and those newer to the field.

In conclusion, by embracing AI platforms like FishSight AI, the industry can advance toward a future where downhole challenges are met with foresight and consistency, embedding the hard-earned lessons of the past into the real-time operations of the present and future.

References

1. Montes, A., Ashok, P., and van Oort, E. (2024). Review of Stuck Pipe Prediction Methods and Future Directions. *SPE Journal* (2025), 1–18. [SPE-220725-PA]
2. Al-Saba, M., and Al-Kaabi, S. (2018). Machine Learning Models for Real-Time Stuck Pipe Prediction Using Surface Drilling Parameters. *SPE Middle East Oil & Gas Show and Conference*. [SPE-189431-MS]
3. Sarwono, A., et al (2022). Stuck Pipe Detection in Geothermal Operation with Support Vector Machine. *Journal of Earth Energy Engineering*, **11**(3), 102–113.
4. Jahanbani, H., et al (2022). Optimized Bottom Hole Assembly Design Using Artificial Neural Networks: A Case Study from the Middle East. *Petroleum Exploration and Development*, **49**(3), 512–520.
5. Zhang, X., and Abdulraheem, A. (2020). Intelligent Mud Logging and Formation Evaluation Using AI-Driven Platforms. *Journal of Natural Gas Science and Engineering*, **78**, 103267.
6. Shadravan, A., et al (2019). Digital Twin Applications in Wellbore Operations: Real-Time Advisory System for Completion and Intervention. *SPE Digital Energy Conference*. [SPE-194367-MS]
7. Zhou, L., et al (2023). Toward a Digital Fishing Toolbox: Data-Driven Approaches to Downhole Equipment Recovery. *SPE Asia Pacific Oil & Gas Conference*. [SPE-213076-MS]
8. Katzmann, L., Mayer, H., Arévalo, P., and Forshaw, M. (2024). Machine Learning Accelerates and Facilitates the Prediction of Stuck Pipe Events. *Abu Dhabi International Petroleum Exhibition & Conference*. [SPE-222438-MS]
9. AitAli, R., Arévalo, P., Dashevskiy, D., Mahmoud, M., Pocaterra, A., Al Bouny, A., Hussain, M. Q., Baruno, A., & Mohammed, S. 2023. Hybrid data driven intelligent algorithm for stuck pipe prevention. *ADIPEC 2023*. [SPE-216325-MS].
10. Alaskar, S., and Alluqmani, M. (2023). Well Interventions: A Reflection on the Past and the Present. *The Way Ahead Magazine (SPE)*, September 2023.
11. Gao, Y., et al (2021). AI-Assisted Drilling Operation Optimization: Case Studies from the Field. *Journal of Petroleum Science and Engineering*, **200**, 108400.
12. Mohammed, T., et al (2020). Artificial Intelligence and Machine Learning Applications in Drilling Optimization: A Review. Society of Petroleum Engineers. [SPE-200134-MS]
13. SPE Drilling and Completions Technical Section. (2024). Guidelines for AI Implementation in Drilling Operations. Society of Petroleum Engineers.